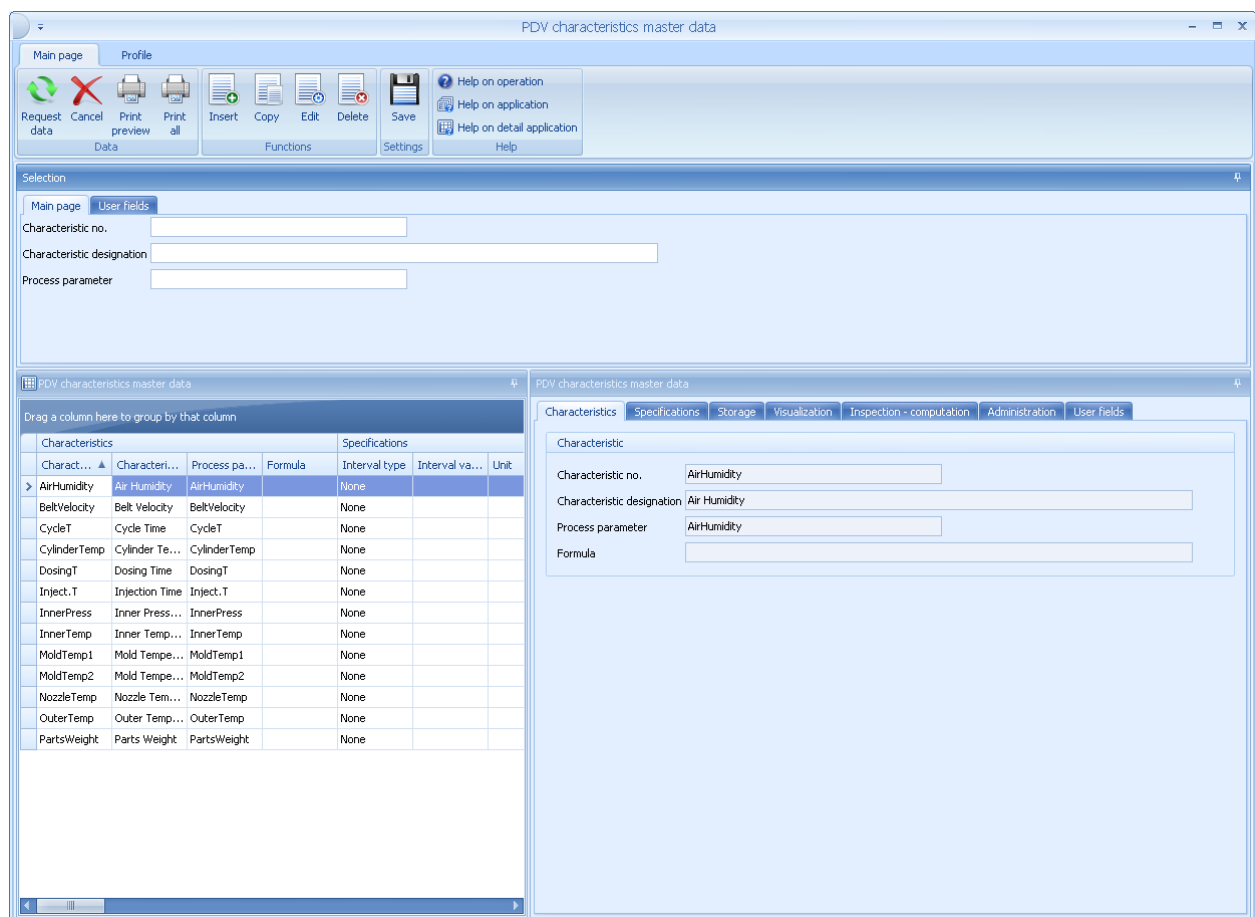


1 PDV Characteristics Master Data

Overview

Menu	Master data → Process data processing → PDV characteristics master data
Transaction code	chrp
Function authorization	chrq.*

You use the catalog of characteristics to predefine characteristics that are then used in collection rules. It aims at persons involved in data collection planning. The characteristics catalog is one of the most important basic catalogs. You need the characteristics catalog to create collection rules. As this catalog is used to predefine characteristics data for collection rules, it includes extensive input options.



Characteristics	Character...	Process pa...	Formula	Interval type	Interval va...	Unit
AirHumidity	Air Humidity	AirHumidity		None		
BeltVelocity	Belt Velocity	BeltVelocity		None		
CycleT	Cycle Time	CycleT		None		
CylinderTemp	Cylinder Te...	CylinderTemp		None		
DosingT	Dosing Time	DosingT		None		
Inject.T	Injection Time	Inject.T		None		
InnerPress	Inner Press...	InnerPress		None		
InnerTemp	Inner Temp...	InnerTemp		None		
MoldTemp1	Mold Tempe...	MoldTemp1		None		
MoldTemp2	Mold Tempe...	MoldTemp2		None		
NozzleTemp	Nozzle Tem...	NozzleTemp		None		
OuterTemp	Outer Temp...	OuterTemp		None		
PartsWeight	Parts Weight	PartsWeight		None		

Drag a column here to group by that column

Characteristic no.

Characteristic designation

Process parameter

Formula

Purpose

The characteristics catalog is one of the most important basic catalogs. You need the characteristics catalog to create collection rules. As this catalog is used to predefine characteristics data for collection rules, it includes extensive input options. Only enter data in the characteristics catalog that need not be changed when later assigned to a collection rule. Do not define limit values, for example. This is normally not useful because the limit values are only known when the collection rule is created. Only when you create and assign a collection rule, a relation to a concrete article or machine is established. Note this and you will know what kind of information you should predefine. For example, it must be carefully considered whether the characteristic "outer diameter" is only created once and detailed information is stored in the collection rule later on or whether several "outer diameter characteristics" are created, e. g. with specification of limit values. Usually, it is an advantage to store a small number of general characteristics. The required evaluations/reports also play a role in this context. If a new "outer diameter characteristic" is created for almost every tolerance change, this characteristic is "valid" for one article or machine only. In a subsequent failure analysis, a comprehensive evaluation is not possible in this case!

It is important that you can still change any definition stored in the catalog or add missing definitions when you later plan the collection rule.

Important

The configurations made in the characteristics' master data need not be definitive for the collection rule. The characteristics' master data is used as a template when the collection rules are later created. You can complete and change all configurations of the characteristics' master data in the collection rule.

Integration

The characteristics' master data is used as a template when the collection rules are later created. The field *Process parameter* provides a logical connection to the logical channels. The connection is only established when the characteristics are definitively defined in the collection rules.

Selection criteria

The application provides the following selection criteria:

- Characteristic no.:
Characteristic number
- Designation:
Name of the characteristic
- Process parameter:
Process parameter

If several selection criteria are used, the overlapping results are displayed below in the *Characteristics master data*.

Field descriptions

Find below a description of the columns and input options for characteristics:

Tab *Characteristics*

Characteristic no.

Unique number of the characteristic

Characteristic designation

Name of the characteristic

Process parameter

Predefinition of the process parameter

Formula

See section "Formula calculation" below.

Tab *Specifications*

Select the *Specifications* tab to enter the sampling scheme and the tolerance limits. Note: As mentioned above, it is only reasonable to define tolerance limits in the characteristics master data under certain circumstances.

Sampling scheme (no longer relevant as of PDV 8.3)

The following sampling schemes are available:

- None
- Piece-related

With a piece-related sampling scheme, you can define the interval. All single values collected in this interval are then combined and form one sample. If you define a piece-related sampling scheme, you can make evaluations for samples, for example control charts.

Interval value (no longer relevant as of PDV 8.3)

Interval used to combine the single values collected in this interval to a sample.

Display of failures in the Failure Mode Analysis (PDV 8.1 and PDV 8.2)



To display the created failures in the Failure Mode Analysis, you must specify an interval value. Process parameters without specified interval value are not integrated in the Failure Mode Analysis.

**Field visibility**

The field is only visible and input is only possible if the user is assigned the function authorization "InspectionInterval".

Unit

Pieces, meter, kg, etc. Units are assigned using the unit catalog.

Decimal places (no longer relevant as of PDV 8.1)

Enter the decimal places. Leading zeros before the comma are not displayed in the specification fields. By default, the number of decimal places defined in the system settings is pre-assigned.

Size (measure type)

Validation and tolerance limits can be entered as absolute, relative or percentage values. Note: You must enter relative or percentage limits as negative values.

Upper PL

Specifies the upper process limit. In PDV, this value also defines the displayed upper red area in process visualization.

Upper TL

Specifies the upper tolerance limit (upper specification limit)

Target value

Specifies the target value

Lower TL

Specifies the lower tolerance limit (lower specification limit)

Lower PL

Specifies the lower process limit. In PDV, this value also defines the displayed lower red area in process visualization.

Upper TL – Generate failure / Lower TL

If the checkboxes *Generate failure* are enabled, a violation of the limit value automatically results (in the background) in the generation of a failure with failure type "limit value violation" (AUTO:TG> or AUTO:TG<) when measured values are collected. The generated failures are evaluated in the Failure Mode Analysis, for example.

Specifications process**Upper PAL**

Upper process action limit

Lower PAL

Lower process action limit

Generate failure

If the checkboxes *Generate failure* are enabled, a violation of the limit value automatically results (in the background) in the generation of a failure with failure type "limit value violation" (AUTO:PEG> or AUTO:PEG<) when measured values are collected. The generated failures are evaluated in the Failure Mode Analysis, for example.

Storage (no longer relevant as of PDV 8.3)

Using the filter functions for storage, you define the frequency used to store single values. This value affects the further processing of measured values on the HYDRA server. In later evaluations, only the actually stored values can be displayed. The storage frequency is also the basis for further aggregations by the PDV Distributor calculating samples. Here, the PDV Distributor only accesses data of the online data set.

Example: The defined filter specifies that only one of ten measured values is saved. At the same time, a sample interval specifies that 5 measured values are combined to one sample. After filtering, the storage frequency is the basis for the calculation of samples. 5 measured values are then actually combined to one sample. Result: For 50 measured values, which have actually been collected, one sample is generated.

Filter function:

Define a filter function for the collection of PDV values. The following four filter functions are available: *None*, *Cyclic*, *Frequency* and *Percentage*. Depending on the selection made, different input fields are shown below the combo box. Use these fields to parameterize the filter function.

- *None*

No filter, the system saves each collected value.

- *Frequency*

Enter a number of measured values. This means that the system only saves every n measured value. The system evaluates the measured values collected in the meantime but they are not saved.

- *Cyclic*

Enter a time interval in seconds. This means that the system only saves a measured value every n seconds. The system evaluates the measured values collected in the meantime but they are not saved.

- *Percentage*

Enter a time interval and a percentage. In the PDV data collection, the system saves the

measured value every n seconds or if the absolute deviation of the current measurement compared to the previous measurement exceeds the percentage entered here. The time interval is reset when the measured value is saved because of the percentage deviation. The system evaluates the measured values collected in the meantime but they are not saved.

Visualization

Visualization

Check this option to enable the online visualization for the process parameter.

Position

Use this integer field to specify the visualization position of the process parameter. Use this for graphics in the online visualization with more than one display element on one page. Example: If 16 display elements are shown on one page, you can use this option to define the position of the different process parameters.

If the value 0 or less is specified, the process parameter is not visualized. You can only edit the input field after you have checked the option *Visualize*.

Visualization *Filter function*

Here, the same filter functions are available as for the storage, but you can use this second setting to decouple the storage from the online visualization. For example, you can configure that each measured value is stored, but only every tenth measured value is visualized.

The following four filter functions are available: *None*, *Cyclic*, *Frequency* and *Percentage*. Depending on the selection made, different input fields are shown below the combo box. Use these fields to parameterize the filter function.

- *None*
No filter, the system visualizes each collected value.
- *Frequency*
Enter a number of measured values. This means that the system only visualizes every n measured value.
- *Cyclic*
Enter a time interval in seconds. This means that the system only visualizes a measured value every n seconds.
- *Percentage*
Enter a time interval and a percentage. In the PDV data collection, the system visualizes a measured value every n seconds or if the absolute deviation of the current measurement compared to the previous measurement exceeds the percentage entered here. The time interval is reset when the measured value is visualized because of the percentage deviation.

Inspection – computation

Check characteristic (no longer relevant as of PDV 8.3)

You can use this option to define whether the process parameter is checked against limit value violations in the PDV data collection or whether the collected measured values only pass the data collection to be stored.

This field also affects the online visualization. You can only display characteristics that are processed by the logic of the PDV data collection and are not only stored.

Formula parameters only

Use this option to define that the created process parameter is only used as parameter to calculate a further process parameter. If you set this option for a process parameter, this process parameter will never assess measured values with respect to violated limit values and these process parameters will never be stored in the database.

Compute limit values

If you enable this option, the fields for the calculation of target values or limits are activated. Formula calculation, see section "Formula calculation"

Upper TL formula

You can use the formula for the calculation of the UTL to calculate a new UTL in combination with other parameters. If you recalculate the UTL, this will also lead to changed target values. See section 3.5 for further information on formulas.

Upper PAL formula

You can use the formula for the calculation of the UPAL to calculate a new UPAL in combination with other parameters. If you recalculate the UPAL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Target value formula

You can use the formula for the calculation of the target value to calculate a new target value in combination with other parameters. If you recalculate the target value, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Lower PAL formula

You can use the formula for the calculation of the LPAL to calculate a new LPAL in combination with other parameters. If you recalculate the LPAL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Lower TL formula

You can use the formula for the calculation of the LTL to calculate a new LTL in combination with other parameters. If you recalculate the LTL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Formula calculation

Using the formula calculation function, you can calculate the measured values collected via machine connection before you evaluate them. For example, you can add an offset to the calculation of a characteristic, you can directly convert units in the collection process or you can perform calculations in combination with other characteristics.



To store and display a formula, the user must have the function authorization `iriscp.formula`. If the user is not authorized, the field is not displayed.

You specify the calculation of measured values in the application *Characteristics* in tab *General*. You can also calculate the target values of an automatically collected characteristic and the upper/lower tolerance or action limits.

Important: HYDRA versions below MES Weaver 2.0 with server systems AIX, HP-UNIX, SCO-UNIX or DEC ALPHA do not provide this kind of formula calculation.

The first part of the formula specifies the level where the formula calculation is made. The following types are available:

- V - Calculation on the level of single values without self-reference.
For each measured value of the characteristics involved, exactly one single value is generated for the calculated characteristic.
- O - Calculation on the level of samples with self-reference.
Use this type of formula to refer to the value itself and to integrate limit or target value.

After the above identifier, the actual formula is specified. The following operators, functions and constants are supported:

Functions	
abs(x)	Calculates the absolute value
atan(x)	Calculates the arc tangent
cosh(x)	Calculates the hyperbolic cosine
float(x)	Converts the value into a floating point number
sqrt(x)	Calculates the square root
acos(x)	Calculates the arc cosine
atan2(y,x)	Calculates the arc tangent of y/x
exp(x)	Calculates the exponential value
log(x)	Calculates the natural logarithm
sin(x)	Calculates the sine
tan(x)	Calculates the tangent
asin(x)	Calculates the arc sine
cos(x)	Calculates the cosine
int(x)	Converts the value into an integer
log10(x)	Calculates the common logarithm
round(x)	Rounds to integer value
round(x,y)	Rounds the value x to y decimal places
sinh(x)	Calculates the hyperbolic sine
tanh(x)	Calculates the hyperbolic tangent
trunc(x)	Reduces the value x to an integer value
trunc(x,y)	Reduces the value x to y decimal places
Operators	
x + y	Addition
x – y	Subtraction
x / y	Division
x * y	Multiplication
x ** y	Calculates x to the power of y
Constants	
pi	3.141592654
e	2.718281828

If constant numeric values are used in formulas, you must be careful not to use thousand separators. If these constants are floating point numbers, be careful to use a dot as decimal separator instead of a comma.

The following syntax [A:B:C]. applies for the variables that identify the single or default values of the process parameters involved. The available values are listed below.

You can specify the following values for section A:

- X – single value/measured value
- UTL – Upper Tolerance Limit
- UPL – Upper Process Action Limit
- TV – Target Value
- LPL – Lower Process Action Limit
- LTL – Lower Tolerance Limit

Note: The value X defined for section A (single value/measured value) is only used for the calculation of measured values; this means, the value is not used in the target value formulas or formulas of the upper/lower tolerance or action limits.

Note: Aggregate functions like MAX, MIN or AVG, which are used in HYDRA CAQ, are not supported with automatically collected characteristics.

Section B describes how the relevant characteristic is identified. The following possibilities are available:

- SELF – self-reference
(This requires formula type O)
- PPARAM – Reference to further process parameters of the same machine
With this formula, only process parameters of the same machine can be calculated.

Section C identifies the characteristic using the field content specified in section B. This means: If a self-reference is specified in section B, a section C may not exist. If instead a reference to a further process parameter is stored for PPARAM, the process parameter must be specified in section C.

Example 1:

The process parameter always collects ten times the rounded value from the machine connection.

→ Formula: O: `round([X:SELF]) * 10`

Example 2:

The characteristic "area" results from the product of the process parameters LENGTH (LAENGE) and WIDTH (BREITE). For each single value of the two source characteristics, a single value is calculated for the characteristic "area".

→ Formula: V: [X:PPARAM:LAENGE] * [X:PPARAM:BREITE]

Example 3:

The following formula is not stored for the measured value, but for the target value of the process parameter "speed". The target value is calculated using the own target value plus the target value of process parameter DURCHMESSER (diameter). The target value for "speed" is therefore recalculated, if the target value for "diameter" changes.

→ Formula: O: [TV:SELF] + [TV:PPARAM:DURCHMESSER]

Note:

The formula calculation is performed directly after reception of the measured values from the PDV data collection. Changes to any of the process parameters included in the formula are then used as events that trigger recalculation.